CHE 565 -- Exam 2

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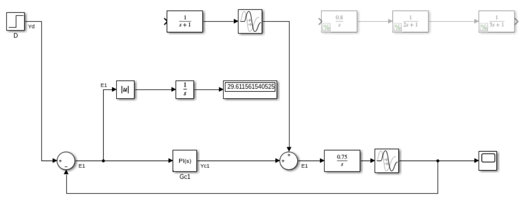
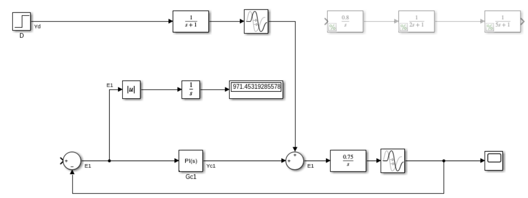
Problem 1

Parts A and B: Basic PI Feedback Control



(a) Setpoint step response



(b) Disturbance step response

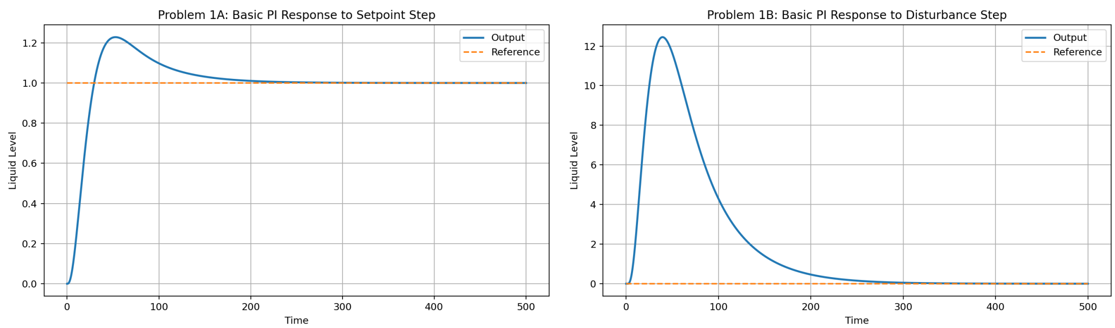
The actual boiler liquid-level process is used for the closed-loop simulation, while the integrating-plus-delay model is used only to obtain the controller settings.

$$
\begin{aligned}
G_{\text{actual}}(s) &= \frac{0.8}{s(2s+1)(5s+1)} \\
G_{\text{model}}(s) &= \frac{0.75}{s}e^{-4s} \\
G_{\text{d}}(s) &= \frac{1}{s+1}e^{-s}
\end{aligned}
$$

Using the lambda-tuning rule with an integrating-plus-delay model:

$$
\begin{aligned}
\lambda &= 8\theta \\
\tau_I &= \max(4\theta,\ 2\lambda+\theta) \\
K' &= \frac{K_p}{\tau_p} \\
K_c &= \frac{\tau_I}{K'(\lambda+\theta)^2}
\end{aligned}
$$

$$\theta = 4.0$$
$$\lambda = 32.0000$$
$$\tau_I = 68.0000$$
$$K_c = 0.0700$$



(a) Setpoint step

(b) Disturbance step

Figure 2: Basic PI closed-loop responses.

The IAE values for the basic PI controller are:

$$IAE_{\text{setpoint,basic}} = 31.9746$$
$$IAE_{\text{disturbance,basic}} = 971.9761$$

For the setpoint response, the error is measured relative to a final desired value of 1. For the disturbance-rejection response, the desired value is 0 because the controller should reject the disturbance and return the output to its original operating point.

Part C: Cascade Control Structure

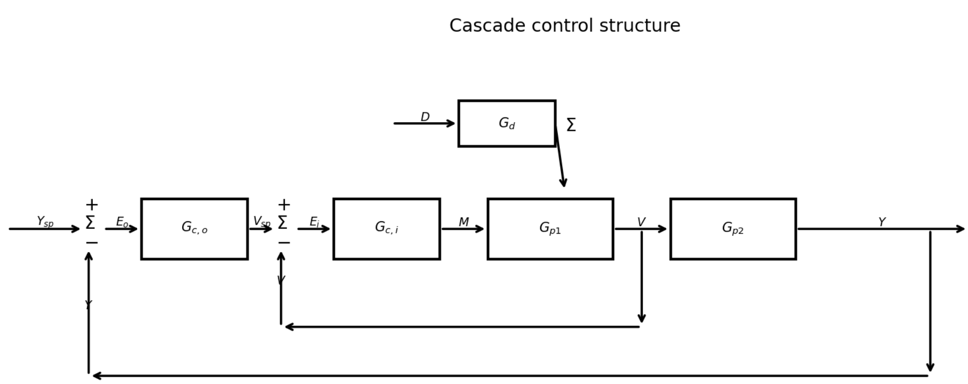
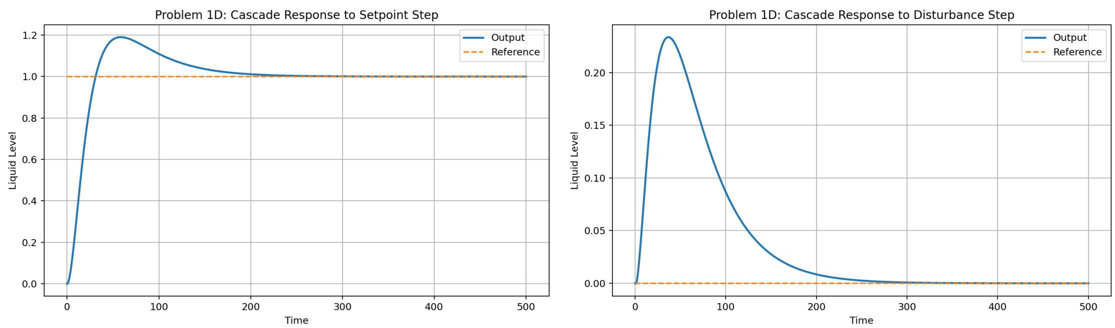


Figure 3: Cascade control diagram.

A cascade loop is added by splitting the process into an inner process and an outer process. The inner controller is P-only, while the outer controller remains PI. The disturbance is assumed to enter the inner-loop process variable, allowing the inner loop to reject the disturbance before it propagates through the slower boiler-level dynamics.

$$G_{p1}(s) = \frac{1}{2s+1}$$
$$G_{p2}(s) = \frac{0.8}{s(5s+1)}$$
$$G_{actual}(s) = G_{p1}(s)G_{p2}(s)$$

Part D: Cascade Control with Inner Gain of 50



(a) Cascade setpoint step

(b) Cascade disturbance step

Figure 4: Cascade closed-loop responses with inner controller gain equal to 50.

The IAE values for the cascade controller are:

$$K_{\text{c,inner}} = 50.0000$$
$$IAE_{\text{setpoint,cascade}} = 30.2838$$
$$IAE_{\text{disturbance,cascade}} = 19.4403$$

Part E: Discussion

The effectiveness of the cascade controller can be evaluated by comparing the IAE values for the basic feedback controller and the cascade controller. A lower IAE indicates better overall control performance because the total accumulated error is smaller.

$$IAE_{\text{setpoint,basic}} = 31.9746$$
$$IAE_{\text{setpoint,cascade}} = 30.2838$$
$$IAE_{\text{disturbance,basic}} = 971.9761$$
$$IAE_{\text{disturbance,cascade}} = 19.4403$$

The cascade controller improves disturbance rejection because the inner loop reacts directly to the disturbance before the disturbance fully propagates through the outer liquid-level dynamics.

Problem 2

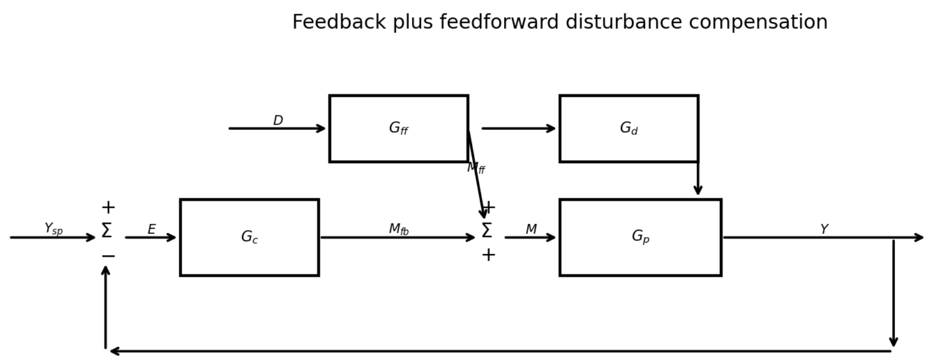


Figure 5: Feedforward plus feedback control diagram.

For the feedforward design, the disturbance is assumed to enter on the plant-input side. The manipulated variable and the disturbance therefore both pass through the same boiler-level process dynamics.

$$\begin{aligned} Y(s) &= G_p(s)\,[M(s) + G_d(s)D(s)] \\ M(s) &= M_{fb}(s) + M_{ff}(s) \\ M_{ff}(s) &= G_{ff}(s)D(s) \end{aligned}$$

For perfect disturbance cancellation, the disturbance contribution inside the plant input must be zero:

$$\begin{aligned} G_{ff}(s)D(s) + G_d(s)D(s) &= 0 \\ G_{ff}(s) &= -G_d(s) \end{aligned}$$

Using the disturbance transfer function:

$$G_d(s) = \frac{1}{s+1}e^{-s}$$
$$\boxed{G_{ff}(s) = -\frac{1}{s+1}e^{-s}}$$

This result follows because the disturbance enters at the same summing point as the manipulated variable before the plant. Therefore, the plant transfer function cancels out in the feedforward design.

Problem 3

Part A: RGA and Pairing

The steady-state gain matrix is found by evaluating each transfer function at s = 0.

$$K = \begin{bmatrix} 2.3 & 0.47 \\ -1.2 & 0.58 \end{bmatrix}$$
$$\Lambda = K \circ (K^{-1})^T$$

$$\Lambda = \begin{bmatrix} 0.7028 & 0.2972 \\ 0.2972 & 0.7028 \end{bmatrix}$$

The diagonal RGA values are closer to 1 than the off-diagonal values, so the appropriate pairings are:

$$Y_1 \leftrightarrow M_1$$
$$Y_2 \leftrightarrow M_2$$

Part B: MIMO Feedback Block Diagram

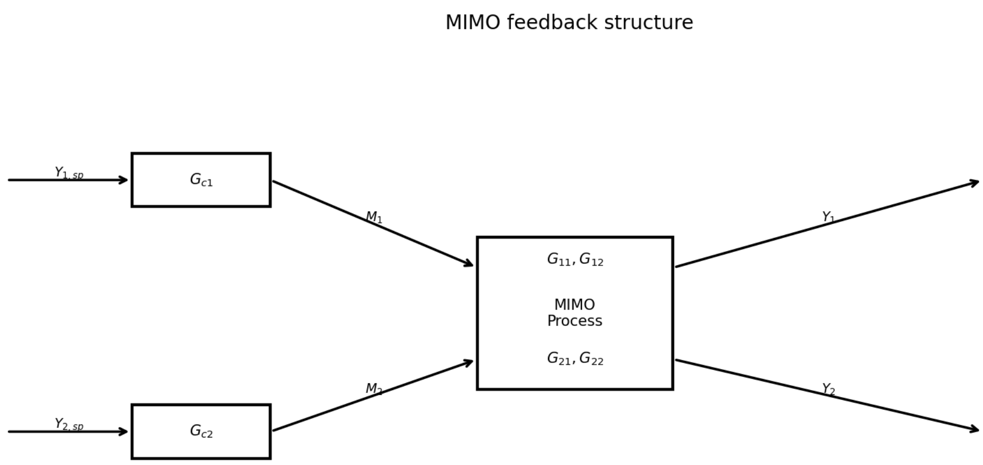


Figure 6: MIMO feedback control diagram without decouplers.

The MIMO process contains four transfer functions. Each manipulated variable affects both output variables, which creates loop interaction.

$$G_{11}(s) = \frac{2.3}{4.6s+1}e^{-0.2s}$$
$$G_{12}(s) = \frac{0.47}{2.2s+1}e^{-0.3s}$$
$$G_{21}(s) = \frac{-1.2}{18s+1}e^{-0.4s}$$
$$G_{22}(s) = \frac{0.58}{1.8s+1}e^{-0.5s}$$

Part C: Decoupler Design

For diagonal pairing, the decouplers are selected to cancel the off-diagonal process interactions.

$$D_{12}(s) = -\frac{G_{12}(s)}{G_{11}(s)}$$
$$D_{21}(s) = -\frac{G_{21}(s)}{G_{22}(s)}$$

Therefore:

$$D_{12}(s) = (-0.2043)\frac{4.6s+1}{2.2s+1}e^{-0.1s}$$
$$D_{21}(s) = (2.0690)\frac{1.8s+1}{18s+1}e^{0.1s}$$

The first decoupler contains an additional delay and is causal. The second decoupler contains a time advance, represented by the positive exponential term. A pure time advance is noncausal, so an exact implementation of this decoupler is not physically realizable. In practice, this decoupler would need to be approximated or modified.

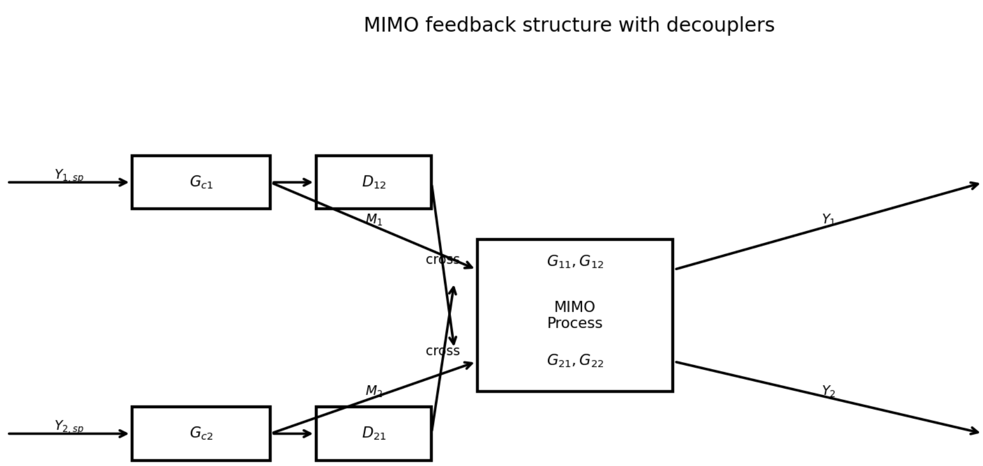


Figure 7: MIMO feedback control diagram with decouplers.